Curves, reverses, and loops

The challenges of building the line to Darjeeling were apparent right from the start. It was not just the terrain, soil erosion, or the seemingly insurmountable task of cutting through the mountains. This was the first time anyone had undertaken such an endeavour – of building a narrow-gauge line over 80 km of unknown terrain. The solution lay in innovative engineering.

The story goes that the engineer assigned to the task faced an unsolvable problem at the town of Tindharia, which happened to be the steepest portion on the line. Erosion on the hills did not allow for building a permissible gradient. Frustrated, he threw up his hands in despair when his wife said, "Darling if you can't go ahead, why don't you come back?" And so six "z-shaped reversing stations", similar to the one built by the Great Indian Peninsula Railway (GIPR) in Bhore Ghat (see Conquering The Ghats), were used on this line. These had the steepest gradient of 1 in 18 and UNESCO noted that the combination of narrow gauge and "zigzag reverses" was the first of its kind in the world.

The line is equally significant for its loops that seem to take the train to the absolute edge of the mountain. Of the five loops, three are still operational and the sharpest, with a radius of about 18 m, is aptly named Agony Point. The famous Batasia Loop, with its view of the snow-capped Kanchenjunga mountain, is the last as the line approaches Ghoom, which at 2,258 m is one of the highest railway station in the world.

The engines

If any locomotive defines a railway line, it is the DHR B Class. For many years these small, yet powerful, engines have hauled trains up the mountain. The first four were built in 1889 by UK-based Sharp, Stewart and Company. By 1927, the North British Locomotive Company of Glasgow, the Baldwin Locomotive Works of Philadelphia in the US, and the Tindharia Works in Bengal built a further 25. Nowadays, some B Class engines continue to run on the DHR, while several serve as retired exhibits in museums around India.